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CSCI 6515 NLU

Formal Language Bias in LLMs

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Overview of the Task & Research Questions

Problem Definition

Large language models often sound more formal than everyday human conversation. We investigate whether this is measurable as **register bias**.

Tasks

Lexicon-based analysis

- Formality Rate (FR)
- Corpus-level comparisons

Fine tuning analysis and mitigation

- RoBERTa register formal vs informal classifier
- LoRA Mitigation Fine-tuning

Research Questions

- Do LLM outputs exhibit higher formal register than human conversational corpora?
- Can lexical and learned metrics find which models show stronger formality tendencies?
- Does the bias originate from pretraining data or RLHF instruction fine-tuning?
- Can fine-tuning on low-formality data reduce bias without degrading fluency?

Motivation I

The "Delve" Problem

LLMs often favor elevated words like *delve*, *utilize*, and *commence* over everyday alternatives like *dig into*, *use*, or *start*. This suggests a systematic stylistic preference toward formal register rather than ordinary conversational language.

Conversational Register Mismatch

When assistants respond in document-style prose rather than natural dialogue, they can sound unnatural, reduce user alignment, and weaken their effectiveness as interactive agents. The formal language sounds distant and unrelatable to a human user.

Real-World Impact

Formality can affect accessibility to information due to comprehension difficulties and reduce trust due to reduced user alignment. This can cause issues in applications in education, mental-health support, and consumer assistants where plain language is often critical.

Motivation II

Response 1

Natural Language Understanding (NLU) is a specialized subdomain of Natural Language Processing concerned with the computational derivation of semantic and pragmatic meaning from unstructured linguistic input. It operationalizes techniques such as intent classification and entity extraction to transform raw text into structured, machine-interpretable representations. NLU underpins intelligent systems like Siri and Google Assistant, enabling them to infer user objectives and generate contextually appropriate responses.

Response 2

Natural Language Understanding (NLU) is a part of Natural Language Processing that helps computers understand the meaning of human language. It figures out what a person wants and pulls out important details from what they say or type. NLU is used in tools like Siri and Google Assistant so they can understand and respond to users.

Contribution

01

Dual Formality Measurement Framework

We created two formality measurements FR and BERT which capture orthogonal dimensions of register. While FR measures directly frequency of formal language, BERT attributes how formal the structure of the response is through pre-training on the GYAFC dataset and fine-tuning on the Pavlick dataset.

02

Cross-Family Bias Evidence and Causes

We created the first large-scale evidence that formality bias is universal across GPT, LLaMA, Mistral, Claude, and Vicuna families. We also found evidence is caused by a combination of conversational topic, fine-tuning, and pre training data.

03

Fine-Tuning Formality Bias Mitigation

We test whether LoRA fine-tuning can shift Qwen2.5-3B-Instruct toward a lower-formality register while preserving basic fluency and coherence.

03

Register Bias Analysis

Because LLMs show elevated formality across several models, FR and RoBERTa scores can help analyze assistant-style register patterns, though they should not be treated as a standalone LLM detector.

Brief Literature Review

Khojah et al. (2024)

Human-Bot Register Differences

AIware

Compared human-human vs human-bot conversations, showing systematic stylistic and register differences in conversational agents.

Pavlick & Tetreault (2016)

Automatic Formality Detection

TACL

Established lexical and feature-based approaches for detecting formality in text, providing the foundation for our **Formality Rate (FR)** metric, and giving us the fine-tuning data for our BERT metric.

Rao & Tetreault (2018) — GYAFC

Style Transfer and Formality Corpora

NAACL

Introduced the GYAFC benchmark for formal/informal rewriting, which is the dataset that our **RoBERTa Register Classifier** metric is pretrained on.

Zhao et al. (2021)

Pretraining and Linguistic Bias

ICML

Showed that pretraining data composition influences systematic model behavior, motivating our hypothesis that register bias may emerge from training dynamics.

Methods I

Datasets & Preprocessing

Corpora

- **LMSYS-Chat-1M** LLM responses across models
- **Pavlick Formality** corpus of sentences and phrases with corresponding formality measurement
- **GYAFC** supervision for formality classification
- **Human Baselines**
 - **Reddit**: Informal web text
 - **ELI5**: Explanatory Q&A
 - **BlendedSkillTalk**: Scripted dialogue

Preprocessing

- Filter conversational turns
- Remove markup / assistant artifacts
- Normalize text and isolate dialogue responses

Formality Rate (FR) Baseline

$$FR = \frac{\text{formal lexical hits}}{\text{tokens}} \times 1000$$

- Two-tiered Lexicon: Classical LLM-isms and Broad Formal Vocab
- Lightweight, interpretable, lexicon-based benchmark
- Measures formal-register frequency per 1k tokens without needing ML
- Uses domain-specific vocabulary that BERT may not weigh heavily

Methods II

RoBERTa Register Classifier

RoBERTa

- Load RoBERTa model (s-nlp/roberta-base-formality-ranker) pretrained on GYAFC
- Strong sentence-level representations
- Captures context beyond lexical markers
- Outperforms BERT for stylistic classification

Reported Metrics

- Fine-tuned on Pavlick Formality Scores (osyvokon/pavlick-formality-scores)
- F1 Score, Recall, Precision, & Accuracy

Qwen2.5-3B-Instruct LoRA Fine-Tuning

Qwen2.5-3B-Instruct

We apply LoRA fine-tuning on Qwen2.5-3B-Instruct to reduce formality bias without degrading fluency or coherence. Qwen2.5-3B-Instruct was chosen because it's small enough to fit on GPU RAM, and similar enough to modern LLMs to serve as a proof-of-concept for the pipeline.

PEFT/LoRA Fine-Tuning

Fine-tuning LLMs can be expensive and risks overwriting the model's language knowledge, so LoRA freezes the original weights, training small adapter matrices.

Evaluation Framework

Lexicon Baseline (FR)

Method 1: Formality Rate Analysis

Dataset: LMSYS + Human Baselines

Metrics:

- Mean FR/ 1k tokens
- % responses with formal markers
- Tail quantiles (P99)

Statistical Test:

Mann-Whitney U
(LLM vs Human bias significance)

“Is formaily bias measurable?”



Neural Classifier

Method 2: RoBERTa Register Classifier

Dataset: GYAFC Pretraining and Pavlick Fine-Tuning

Metrics:

- Precision
- Recall
- F1
- Accuracy

Validation:

Cross-model scoring + family comparisons

“Does neural evidence support lexical evidence?”



LoRA Mitigation

Method 3: Mitigation Evaluation

Dataset: 50 held-out LMSYS prompts

Metrics:

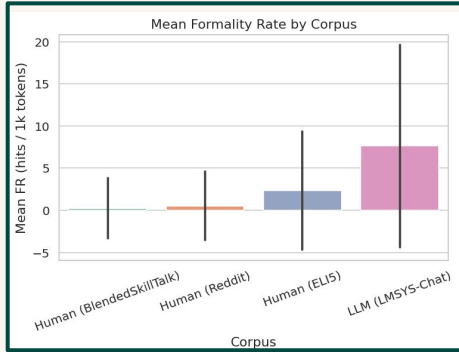
- FR Δ
- BERT Formality Δ
- ROUGE-L

Statistical Test:

Paired t-test
(mitigation effect significance)

“Can bias be reduced?”

Results I: LLM vs Human Baseline Formality



Conversation Type (LMSYS)	FR	BERT P(Formal)
Casual	2.4471	0.6143
General	5.2536	0.7631
Task	4.6810	0.8890

Human Formality

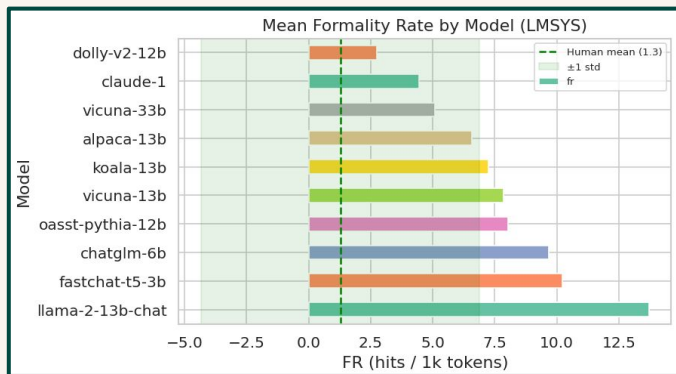
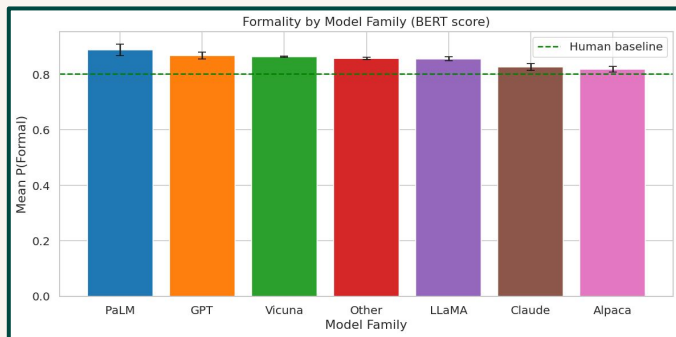
- Reddit and BlendSkillTalk are the most informal probably due to having casual dialogue, while ELI5 is more formal most likely due to Q&A explanatory context

Systematic Bias Found in LLMs

- LMSYS LLMs surpass all human baseline bias significantly
- Conversation type strongly affects formality; casual LMSYS responses are lower than task/general responses but still above casual human baselines.

Baseline and LLM	Mean FR	FR Standard Deviation	Mean BERT P(Formal)	Bert Standard Deviation
BlendedSkillTalk	0.24	3.60	0.271	0.337
Reddit	0.52	4.07	0.245	0.325
ELI5	2.33	7.03	0.801	0.271
LLM (LMSYS-Chat)	7.62	12.02	0.859	0.271

Results II: Formality Bias Across Models



Key Findings

Bias is widespread across model families

- Nearly all families exceed the human formality baseline

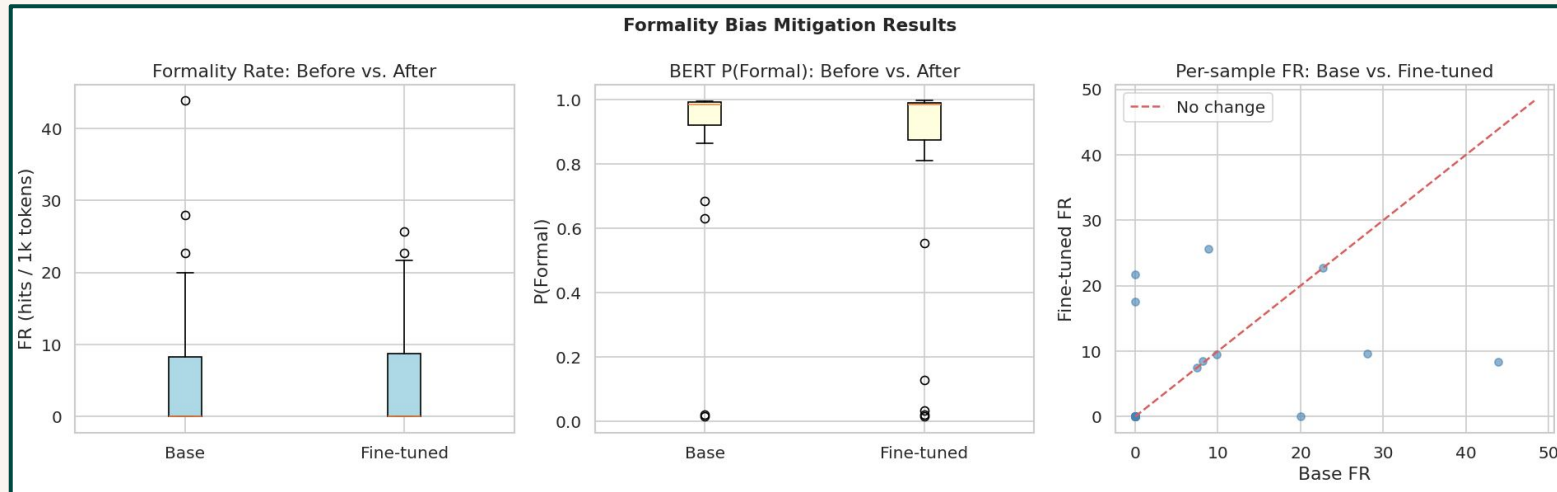
Bias strength varies by family and model

- PaLM and GPT show strongest formal-register tendencies
- Claude is lower, but still near/above human levels

Variation suggests training matters

- Differences across families and individual models point to effects of pretraining and alignment

Results III: LoRA Qwen2.5-3B-Instruct Fine-Tuning



Mean formality decreased

- FR decreased from **6.20** → **5.46**, a **12.0% reduction**
- RoBERTa P(Formal) decreased from **0.868** → **0.794**

Content similarity did not drop

- ROUGE-L changed slightly from **0.0727** → **0.0766**
- This suggests no obvious loss under this metric

Result is preliminary

- The held-out evaluation set is small
- The paired test was not statistically significant
- Mitigation suggests possible steerability, not definitive debiasing

Discussion & Analysis I

FR & BERT

ELI5 represents explanatory QA, so higher formality is expected:

- BERT P(formal) = 0.801
- FR = 2.33

This shows that the two metrics capture different parts of register. While FR measures frequency of formal vocabulary being used, BERT score measures formality classification probability.

Topic-Dependence Formality

Conversation types have a massive impact on formality.

- The formality rate is half the average rate for LLMs
- Casual LMSYS has lower RoBERTa formality than ELI5, but remains above casual human baselines.
- Task responses have the highest RoBERTa P(formal), while general responses have the highest FR.

Top LLM Formal Words

=== Top 20 formal words in LMSYS-Chat ===	
provide	3858
such as	2919
additionally	900
therefore	484
in conclusion	322
implement	303
framework	298
comprehensive	266
prioritize	220
in order to	220
crucial	208
navigate	199
strive	173
enhance	160
indicate	155
obtain	123
optimal	91
robust	80
intricate	74
demonstrate	69
=== Top 20 formal words in Human (ELI5) ===	
such as	256
thus	192
therefore	159
in order to	134
provide	93
hence	62
additionally	35
bandwidth	23
furthermore	21
prior	20
indicate	18
implement	15
optimal	13
subsequent	11
utilize	11
obtain	10
narrative	10
substantial	9
demonstrate	9
navigate	8

Discussion & Analysis II

Universal LLM Formality Bias

Nearly all model families exceed human conversational baselines.

- PaLM/GPT show strongest formal tendencies
- Claude has the lowest but still elevated compared to human baselines

This suggests bias emerges from **shared pre training corpora**, rather than one architecture alone. RLHF may further reinforce a formality tendency across aligned models.

Formality bias appears systemic, not model-specific.

LoRA Shows Partial Register Control

Fine-tuning shifts mean FR and mean RoBERTa formality downward, suggesting that register behavior may be partially steerable through lightweight adaptation.

However, the held-out prompt set is small and the paired test was not statistically significant, so this should be treated as preliminary evidence rather than definitive mitigation.

Future work should test stronger models and include human evaluation of naturalness, clarity, and preference.

Lessons Learned

1

Complementary Metrics Were Necessary

The project reinforced that lexical and neural measures capture different dimensions of register. Formality Rate reflected surface lexical choices, while RoBERTa captured broader stylistic structure.

2

Baseline Selection Shaped Interpretation

Comparisons against multiple human corpora revealed that perceived LLM formality bias depends substantially on conversational genre and reference baseline.

3

Register Mitigation Needs Stronger Evaluation

The LoRA experiment suggested that lightweight adaptation can influence register, but the small held-out set and non-significant paired test mean the result should be treated as preliminary.

4

Data Preparation Influenced the Results

Preprocessing decisions—particularly filtering artifacts, multilingual content, and formatting noise—had measurable effects on both lexical and neural analyses.

Limitations & Future Work

Limitations

Measurement limitations

- FR and RoBERTa are proxy metrics; neither fully captures perceived conversational naturalness
- GYAFC-trained classifier may not generalize to all conversational register styles

Causal limitation

- Pretraining vs. RLHF effects remain confounded by model family and data mix
- Cross-model comparisons suggest sources of bias, but do not isolate causality

Mitigation limitations

- Qwen2.5-3B-Instruct LoRA is still proof-of-concept; the held-out evaluation set is too small for definitive mitigation claims
- Paired testing did not show a statistically significant reduction, so results are preliminary

Future Work

Scale mitigation

- Apply LoRA / DPO to Minstral 7B and LLaMA-3-8B for stronger register control

Human-centered evaluation

- Run user studies measuring naturalness, trust, and preference before/after mitigation

Causal bias analysis

- Isolate contributions of pretraining, RLHF, and task alignment to formality bias

Extend the framework

- Multilingual register analysis + multi-turn formality drift modeling
- Dynamic lexicon expansion from LLM vs human corpus contrasts

Team Contributions

Oliver Krisetya

Lexicon Construction

Designed and curated the two-tier formal-register lexicon (Tier-1 LLM-isms + Tier-2 broad formal vocab) with informal substitution pairs

Formality Rate Pipeline

Implemented the FR computation pipeline (tokenization, phrase matching, per-source aggregation) and produced all FR visualizations

Bias Source Analysis

Designed the conversation-type classifier (task / casual general) and conducted model-family grouping analysis

Statistical Testing

Ran Mann-Whitney U, Kruskal-Wallis, and paired t-tests; diagnosed zero-inflation issue in BlendedSkillTalk dsitribution

Warren Nguyen

Data Research, Collection, & Preprocessing

Set up HuggingFace streaming pipelines for LMSYS-Chat-1M, BlendedSkillTalk, Reddit, and ELI5; implemented markdown cleaning and length filtering

RoBERTa Classifier

Fine-tuned RoBERTa-base on GYAFC formality scores; hyperparameter search, train/val/test/ splitting, classification report analysis

LoRA Migitation

Implemented PEFT LoRA fine-tuning pipeline using TRL SFTTrainer on Qwen2.5-3B-Instruct; configured low-formality SFT dataset construction

Evaluation & Reporting

Computed ROGUE-L evaluation, qualitative error analysis on 50 pairs, final results CSV export, and notebook organization

Class Name



Student Name

Questions?